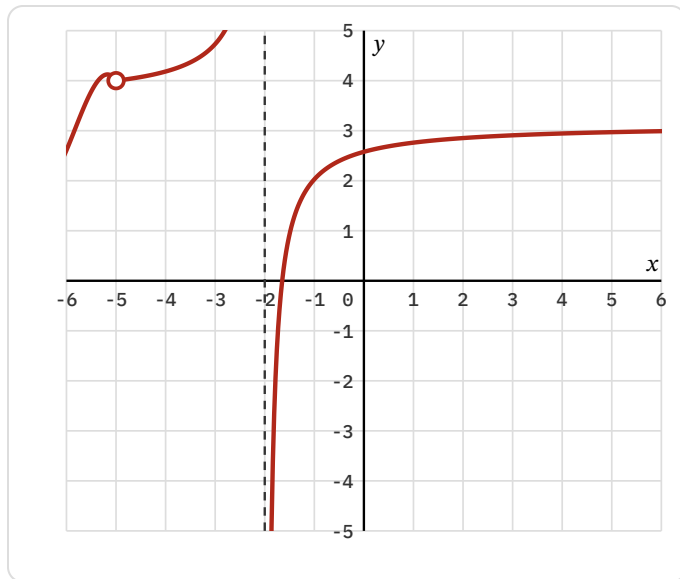


Mock Midterm D — Answer Key

Practice midterm (AI-Free) · Sections 1.1–1.4

Total: 92 points

1. Use the graph of $y = f(x)$ to state the value of each quantity, if it exists. If it does not exist, write DNE. 8 pts



- a.) $f(-5)$ (1)
DNE
- b.) $\lim_{x \rightarrow -5^-} f(x)$ (1)
4
- c.) $\lim_{x \rightarrow -5^+} f(x)$ (1)
4
- d.) $\lim_{x \rightarrow -5} f(x)$ (1)
4
- e.) $f(-2)$ (1)
DNE
- f.) $\lim_{x \rightarrow -2^-} f(x)$ (1)
 ∞
- g.) $\lim_{x \rightarrow -2^+} f(x)$ (1)
 $-\infty$
- h.) $\lim_{x \rightarrow -2} f(x)$ (1)
DNE

WORKED SOLUTION

At $x = -5$

There is no filled dot on the line $x = -5$, so $f(-5)$ is undefined $\rightarrow$ DNE.

Tracing the curve toward $x = -5$ from the left, the heights approach 4 (open or closed dot doesn't matter).

Tracing the curve toward $x = -5$ from the right, the heights approach 4 (open or closed dot doesn't matter).

Both one-sided limits equal 4, so $\lim_{x \rightarrow -5} f(x) = 4$ — even though $f(-5)$ is undefined. The limit ignores the value AT the point.

At $x = -2$

There is no filled dot on the line $x = -2$, so $f(-2)$ is undefined $\rightarrow$ DNE.

From the left, the curve runs along the asymptote $x = -2$ upward, so the limit is ∞ .

From the right, the curve runs along the asymptote $x = -2$ downward, so the limit is $-\infty$.

The left and right limits don't match, so $\lim_{x \rightarrow -2} f(x)$ DNE.

2. Let $f(x) = \frac{-3}{x-1}$. Answer with a number, ∞ , $-\infty$, or DNE.

5 pts

a.) $\lim_{x \rightarrow 1^-} f(x)$ (1)

∞

b.) $\lim_{x \rightarrow 1^+} f(x)$ (1)

$-\infty$

c.) $\lim_{x \rightarrow 1} f(x)$ (1)

DNE

d.) $\lim_{x \rightarrow \infty} f(x)$ (1)

0

e.) The vertical asymptote is $x =$ (1)

1

WORKED SOLUTION

Near $x = 1$ the denominator $x - 1$ is tiny, so $f(x)$ is huge in size. Only the sign needs work.

Left of 1 (try $x = 0.99$): $x - 1$ is a small **negative** number, so $\frac{-3}{\text{small negative}} \rightarrow \infty$.

Right of 1 (try $x = 1.01$): $x - 1$ is a small **positive** number $\rightarrow -\infty$.

The sides disagree, so the two-sided limit DNE.

As $x \rightarrow \infty$, the denominator grows without bound while the top stays -3, so $f(x) \rightarrow 0$. The vertical asymptote is $x = 1$.

3. Calculate each limit algebraically.

6 pts

a.) $\lim_{x \rightarrow 4} (-x^2 - 2x - 5)$ (2)
 -29

b.) $\lim_{x \rightarrow 0} \frac{x^2+9}{x-2}$ (2)
 $-\frac{9}{2}$

c.) $\lim_{x \rightarrow 1} \sqrt{x^2 + 8}$ (2)
 3

WORKED SOLUTION

Each input is in the domain of its function, so direct substitution works.

a. $\lim_{x \rightarrow 4} (-x^2 - 2x - 5) = -1(4)^2 - 2(4) - 5 = -29$

b. $\lim_{x \rightarrow 0} \frac{x^2+9}{x-2} = \frac{(0)^2+0(0)+9}{0-2} = \frac{9}{-2} = -\frac{9}{2}$

c. $\lim_{x \rightarrow 1} \sqrt{x^2 + 8} = \sqrt{(1)^2 + 8} = \sqrt{9} = 3$

4. Calculate each limit algebraically. (Direct substitution gives $\frac{0}{0}$ — simplify first.) If it doesn't exist, write DNE. 4 pts

a.) $\lim_{x \rightarrow 0} \frac{-5x^2-9x}{3x}$ (2)
 -3

b.) $\lim_{x \rightarrow 4} \frac{\sqrt{x}-2}{x-4}$ (2)
 $\frac{1}{4}$

WORKED SOLUTION

a.

$$\lim_{x \rightarrow 0} \frac{-5x^2 - 9x}{3x} = \lim_{x \rightarrow 0} \frac{x(-5x - 9)}{3x} = \lim_{x \rightarrow 0} \frac{-5x - 9}{3} = -3$$

b.

$$\lim_{x \rightarrow 4} \frac{\sqrt{x} - 2}{x - 4} = \lim_{x \rightarrow 4} \frac{\sqrt{x} - 2}{(\sqrt{x} - 2)(\sqrt{x} + 2)} = \lim_{x \rightarrow 4} \frac{1}{\sqrt{x} + 2} = \frac{1}{2 + 2} = \frac{1}{4}$$

Notation tip: keep writing **lim** on every line until the step where you actually substitute.

5. Compute each limit algebraically. Use $\infty / -\infty$ for unbounded results.

6 pts

a.) $\lim_{x \rightarrow \infty} \frac{-2}{5x^4 + 5x^3}$ (2)
0

b.) $\lim_{x \rightarrow -\infty} \frac{2x^2 - 8x + 7}{5x}$ (2)
 $-\infty$

c.) $\lim_{x \rightarrow \infty} \frac{-5x^4 + 6x^3}{6x^4 - 6x^2 - 1}$ (2)
 $-\frac{5}{6}$

WORKED SOLUTION

a. Divide every term by x^4 , the highest power of x in the denominator:

$$\lim_{x \rightarrow \infty} \frac{-2}{5x^4 + 5x^3} = \lim_{x \rightarrow \infty} \frac{-\frac{2}{x^4}}{5 + \frac{5}{x}}$$

The top $\rightarrow 0$ while the bottom $\rightarrow 5$, so the limit is 0.

b. Divide every term by x , the highest power of x in the denominator:

$$\lim_{x \rightarrow -\infty} \frac{2x^2 - 8x + 7}{5x} = \lim_{x \rightarrow -\infty} \frac{2x - 8 + \frac{7}{x}}{5}$$

The bottom $\rightarrow 5$, but the top is dominated by $2x^1$, which $\rightarrow -\infty$ as $x \rightarrow -\infty$; dividing by 5 gives $-\infty$.

c. Divide every term by x^4 , the highest power of x in the denominator:

$$\lim_{x \rightarrow \infty} \frac{-5x^4 + 6x^3}{6x^4 - 6x^2 - 1} = \lim_{x \rightarrow \infty} \frac{-5 + \frac{6}{x}}{6 - \frac{6}{x^2} - \frac{1}{x^4}}$$

Every fraction term $\rightarrow 0$, leaving $\frac{-5}{6} = -\frac{5}{6}$.

6. Find each limit (end behavior). Use ∞ or $-\infty$ when the outputs grow without bound.

4 pts

a.) $\lim_{x \rightarrow -\infty} (x^2 - 5)$ (1)
 ∞

b.) $\lim_{x \rightarrow \infty} (-1 - 3e^{-x})$ (1)
 -1

c.) $\lim_{x \rightarrow -\infty} (-6 - \frac{8}{x^2})$ (1)
 -6

d.) $\lim_{x \rightarrow \infty} (-3x^4 - 5x + 4)$ (1)
 $-\infty$

WORKED SOLUTION

a. Leading term x^2 : degree 2 (even), leading coefficient positive. As $x \rightarrow -\infty$, x^2 is positive, times 1 $\rightarrow \infty$.

b. As $x \rightarrow \infty$, e^{-x} shrinks to 0, so the expression approaches -1.

c. $\frac{8}{x^2} \rightarrow 0$ as $x \rightarrow -\infty$, leaving -6.

d. Leading term $-3x^4$: degree 4 (even), leading coefficient negative. As $x \rightarrow \infty$, x^4 is positive, times -3 $\rightarrow -\infty$.

7. Let

6 pts

$$f(x) = \begin{cases} \frac{x^2 - 4x + 3}{x - 3} & x < 3 \\ 4x - 10 & x > 3 \end{cases}$$

Find each limit or function value algebraically. Write DNE if it doesn't exist.

a.) $\lim_{x \rightarrow 6} f(x)$ (1)
14

b.) $\lim_{x \rightarrow 0} f(x)$ (1)
-1

c.) $f(3)$ (1)
DNE

d.) $\lim_{x \rightarrow 3^-} f(x)$ (1)
2

e.) $\lim_{x \rightarrow 3^+} f(x)$ (1)
2

f.) $\lim_{x \rightarrow 3} f(x)$ (1)
2

WORKED SOLUTION

For $x < 3$, factor the top: $\frac{(x-3)(x-1)}{x-3} = x - 1$ (valid since $x \neq 3$ there).

a. $6 > 3 \rightarrow$ bottom piece: $4(6) - 10 = 14$.

b. $0 < 3 \rightarrow$ top piece, simplified: $0 - 1 = -1$.

c. Neither piece includes $x = 3$, so $f(3)$ is undefined $\rightarrow$ DNE.

d. Left: $\lim_{x \rightarrow 3^-} (x - 1) = 2$.

e. Right: $\lim_{x \rightarrow 3^+} (4x - 10) = 2$.

f. Both sides equal **2**, so the limit is **2**.

8. Is $f(x)$ continuous at $x = -2$?

5 pts

$$f(x) = \begin{cases} \frac{x^2 + x - 2}{x + 2} & x \neq -2 \\ -3 & x = -2 \end{cases}$$

a.) $f(-2)$ (2)

-3

b.) $\lim_{x \rightarrow -2} f(x)$ (2)

-3

c.) Is f continuous at $x = -2$? (1)

(A) Yes — $f(-2)$ exists, the limit exists, and they are equal

WORKED SOLUTION

1. $f(-2) = -3$ (given directly by the second piece).

2. $\lim_{x \rightarrow -2} \frac{(x+2)(x-1)}{(x+2)} = \lim_{x \rightarrow -2} (x-1) = -3$.

3. $-3 = -3$. All three conditions hold → **continuous**.

9. Classify each function at the given x -value.

3 pts

a.) $f(x) = \frac{x^2-25}{x+5}$ at $x = -5$ (1)

(B) Removable discontinuity

b.) $f(x) = \frac{x-2}{x^2-3x+2}$ at $x = 1$ (1)

(D) Infinite discontinuity

c.) $f(x) = \frac{x+1}{x^2-2x-3}$ at $x = -1$ (1)

(B) Removable discontinuity

WORKED SOLUTION

a. $0/0$ at -5 , but it cancels to $x - 5$, so the limit exists (-10) while $f(-5)$ is undefined → removable.

b. Factor: $\frac{(x-2)}{(x-2)(x-1)}$. The $(x-1)$ factor does NOT cancel, so $x = 1$ is a vertical asymptote → infinite.

c. Factor: $\frac{(x+1)}{(x+1)(x-3)}$. The $(x+1)$ cancels, leaving $\frac{1}{x-3} \rightarrow -\frac{1}{4} \rightarrow$ removable.

10. Let $V(t) = t^2 - 30t + 600$, where $V(t)$ represents the number of gallons of water in a tank t minutes after a drain is opened. Find the average rate of change over $[8, 11]$. 7 pts

a.) $V(8)$ (2)

424

b.) $V(11)$ (2)

391

c.) Average rate of change (number only) (2)

-11

d.) Which sentence correctly states the result **with units**? (1)

(D) Between $t = 8$ and $t = 11$ minutes, the amount of water in the tank changes at an average rate of -11 gallons per minute.

WORKED SOLUTION

$$V(8) = 424, \quad V(11) = 391$$

$$\frac{V(11) - V(8)}{11 - 8} = \frac{391 - (424)}{3} = \frac{-33}{3} = -11 \text{ gallons per minute}$$

It's an *average* over an interval (not an instant), and the units are gallons per minute — gallons per minute. The total change would be -33 gallons.

11. Let $f(x) = \frac{1}{x}$. Calculate the average rate of change of f over each interval (rounded answers to 2+ decimals are fine). 5 pts

a.) $[2, 5]$ (1)
 $-\frac{1}{10}$

b.) $[2, 3]$ (1)
 $-\frac{1}{6}$

c.) $[2, 2.5]$ (1)
 $-\frac{1}{5}$

d.) $[2, 2.05]$ (1)
 $-\frac{10}{41}$

e.) As the interval shrinks toward $x = 2$, these values approach... (1)
(A) $-\frac{1}{4}$

WORKED SOLUTION

$$[2, 5] : \frac{\frac{1}{5} - \frac{1}{2}}{3} = -\frac{1}{10} \approx -0.1$$

$$[2, 3] : \frac{\frac{1}{3} - \frac{1}{2}}{1} = -\frac{1}{6} \approx -0.1667$$

$$[2, 2.5] : \frac{\frac{2}{5} - \frac{1}{2}}{0.5} = -\frac{1}{5} \approx -0.2$$

$$[2, 2.05] : \frac{\frac{20}{41} - \frac{1}{2}}{0.05} = -\frac{10}{41} \approx -0.2439$$

Each value is the slope of a secant line through $(2, \frac{1}{2})$. As the second point slides toward the first, the secant slopes approach the slope of the **tangent line**:

$$-\frac{1}{2^2} = -\frac{1}{4} \text{ — the instantaneous rate of change at } x = 2.$$

12. Let $f(x) = \frac{4}{x}$. Find and simplify the difference quotient $\frac{f(x+h)-f(x)}{h}$, then use it to fill in the table. 7 pts

a.) Simplified difference quotient (in terms of x and h) (3)

$$\frac{-4}{x(x+h)}$$

b.) $x = 2, h = 1$ (1)

$$-\frac{2}{3}$$

c.) $x = 2, h = 0.5$ (1)

$$-\frac{4}{5}$$

d.) $x = 2, h = 0.05$ (1)

$$-\frac{40}{41}$$

e.) As $h \rightarrow 0$, what value do these approach? (1)

$$-1$$

WORKED SOLUTION

$$\begin{aligned} \frac{f(x+h)-f(x)}{h} &= \frac{\frac{4}{x+h} - \frac{4}{x}}{h} = \frac{\frac{4x-4(x+h)}{x(x+h)}}{h} \\ &= \frac{-4h}{x(x+h)} \cdot \frac{1}{h} = \frac{-4}{x(x+h)} \end{aligned}$$

Now substitute:

$$x = 2, h = 1 : \quad -\frac{2}{3} = -0.6667$$

$$x = 2, h = 0.5 : \quad -\frac{4}{5} = -0.8$$

$$x = 2, h = 0.05 : \quad -\frac{40}{41} = -0.9756$$

Letting $h \rightarrow 0$ leaves -1 : the instantaneous rate of change (slope of the tangent line) at $x = 2$. That's exactly what section 1.4 formalizes.

13. Let $f(x) = x^2 - 6x - 2$. Use the limit definition $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$ to find $f'(x)$. 8 pts

a.) $f'(x) =$ (3)
 $2x - 6$

b.) $f'(-1) =$ (2)
 -8

c.) Slope of the tangent line to f at $x = 2$ (2)
 -2

d.) What does $f'(-1) = -8$ mean? (1)
(D) The instantaneous rate of change of f at $x = -1$ is -8 — the slope of the tangent line there.

WORKED SOLUTION

$$\begin{aligned} f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{[(x+h)^2 - 6(x+h) - 2] - (x^2 - 6x - 2)}{h} \\ &= \lim_{h \rightarrow 0} \frac{x^2 + 2xh + h^2 - 6x - 6h - 2 - x^2 + 6x + 2}{h} \\ &= \lim_{h \rightarrow 0} \frac{2xh + h^2 - 6h}{h} = \lim_{h \rightarrow 0} \frac{h(2x + h - 6)}{h} \\ &= \lim_{h \rightarrow 0} (2x + h - 6) = 2x - 6 \end{aligned}$$

Then $f'(-1) = -8$ and the slope of the tangent at $x = 2$ is $f'(2) = -2$.

$f'(-1)$ is the *instantaneous* rate of change at one input — the slope of the tangent line, not the height of the graph and not an average over an interval.

14. Let $f(x) = 3x^2 - 3x + 3$. Find the equation of the tangent line to the graph of f at the point $(-2, 21)$. 8 pts

a.) $f'(x) =$ (3)
 $6x - 3$

b.) Slope at $x = -2$ (2)
 -15

c.) Tangent line (3)
 $y = -15x - 9$

WORKED SOLUTION

$$\begin{aligned} f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{[3(x+h)^2 - 3(x+h) + 3] - (3x^2 - 3x + 3)}{h} \\ &= \lim_{h \rightarrow 0} \frac{3x^2 + 6xh + 3h^2 - 3x - 3h + 3 - 3x^2 + 3x - 3}{h} \\ &= \lim_{h \rightarrow 0} \frac{6xh + 3h^2 - 3h}{h} = \lim_{h \rightarrow 0} \frac{h(6x + 3h - 3)}{h} \\ &= \lim_{h \rightarrow 0} (6x + 3h - 3) = 6x - 3 \end{aligned}$$

Slope: $m = f'(-2) = -15$.

$$y - 21 = -15(x + 2) \Rightarrow y = -15x - 9$$

Answers in point-slope form are accepted too.

15. Let $f(x) = x^2 - 4$. Complete the limit definition, then interpret $f'(3)$.

6 pts

$$\lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{((x+h)^2 - 4) - (x^2 - 4)}{h} = ?$$

a.) $f'(x) =$ (3)

$2x$

b.) $f'(3)$ (2)

6

c.) What does $f'(3) = 6$ tell you? (1)

(B) The slope of the tangent line to the graph of f at $x = 3$ is 6.

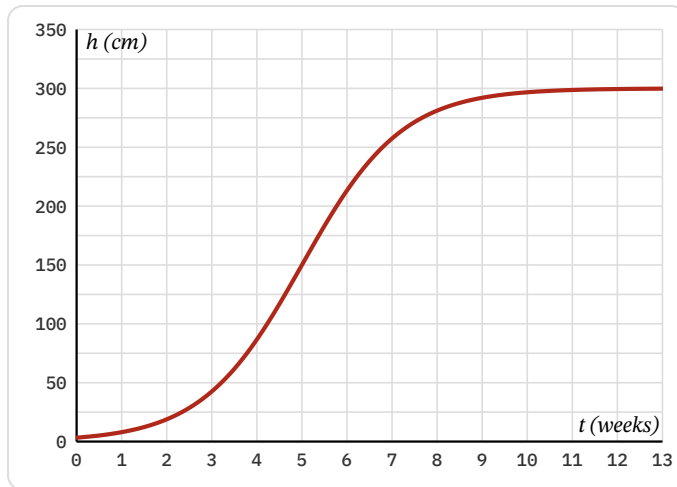
WORKED SOLUTION

$$\lim_{h \rightarrow 0} \frac{x^2 + 2xh + h^2 - 4 - x^2 + 4}{h} = \lim_{h \rightarrow 0} \frac{h(2x + h)}{h} = \lim_{h \rightarrow 0} (2x + h) = 2x$$

$f'(3) = 2(3) = 6$. The graph of f has an instantaneous rate of change of 6 at $x = 3$:
the tangent line there has slope 6 (f is increasing).

16. A sunflower seedling is planted and its height h (in centimeters) after t weeks is graphed below. Decide whether each statement is true or false.

4 pts



- a.) The slopes of the tangent lines get steeper up to $t = 2$, and then the slopes start to flatten out. (1)
False
- b.) The average rate of change over $[0, 12]$ is greater than the instantaneous rate of change at $t = 6$. (1)
False
- c.) The seedling grows faster (instantaneous rate) at week 1 than at week 12. (1)
True
- d.) The average rate of change over $[9, 11]$ is less than over $[4, 6]$. (1)
True

WORKED SOLUTION

a. False. The graph is steepest at about $t = 5$ (where it changes from bending up to bending down).

b. False. $\text{ARC} \approx \frac{299.5 - 3.3}{12} \approx 24.7$ cm/week; the tangent slope at $t = 6$ is about 55.5 cm/week.

c. True. Tangent slopes: about 7 cm/week at $t = 1$ vs. 0.5 at $t = 12$.

d. True. $[9, 11]$: ≈ 3.3 cm/week; $[4, 6]$: ≈ 63.3 cm/week.
